

Cable simulation: which approach to adopt

A decision brief — compatibility first

The question. Three methods were built and checked against a real photographed cable. All reproduce its shape to within a few centimetres, so accuracy does *not* decide this. **Compatibility does** — which existing stack each one plugs into — because the cable is one component of a much bigger project.

On the accuracy numbers. The method closest to the textbook mathematics is the *furthest* from the real cable, and vice-versa. No solver is wrong: the real cable is taped, and tape fixes its angle, which the ideal curve does not model. They differ in *which* real detail they capture — so choose on integration.

Approach	vs real	speed	Integrates with
PhysX capsule chain	33 mm	29 s	Isaac Sim / Omniverse natively: USD, Isaac Lab, ROS 2 bridge, sensors, collision with robots. Real-time.
<i>Best when the cable lives in an Isaac Sim scene and touches things. Watch: rigid capsules; not differentiable.</i>			
Newton cable (add_rod)	75 mm	13 s	The Newton / Warp stack: differentiable and GPU-batched. Newton is the forward-looking engine on the Isaac roadmap.
<i>Best when gradients, learning, or best-in-class cable physics are needed. Watch: evolving API; slowest here.</i>			
Warp XPBD rod	51 mm	1 s	Nothing – it is a self-contained Warp component. Differentiable, portable, drops into any custom pipeline.
<i>Best when a light, embeddable rod is wanted without an engine. Watch: you must calibrate its stiffness and resolution.</i>			

Recommendation.

- Project lives in **Isaac Sim / Omniverse**, cable interacts with robots or the scene → **PhysX capsule chain**. Most compatible, closest to the real cable, real-time.
- Project needs **gradients or GPU-batched learning**, or is adopting Newton → **Newton cable**. Best physics, future-facing, slower.
- Project wants a **light, engine-free** component → **Warp rod**. Fewest dependencies, but you calibrate it.

Ruled out. A Newton *USD-engine* cable cannot be fixed at both ends (an articulation must be a tree; a doubly-clamped cable is a loop), and PhysX *volumetric FEM* builds but its state cannot be read back on this build. Both fail with a clear diagnosis. The three above are the live options.

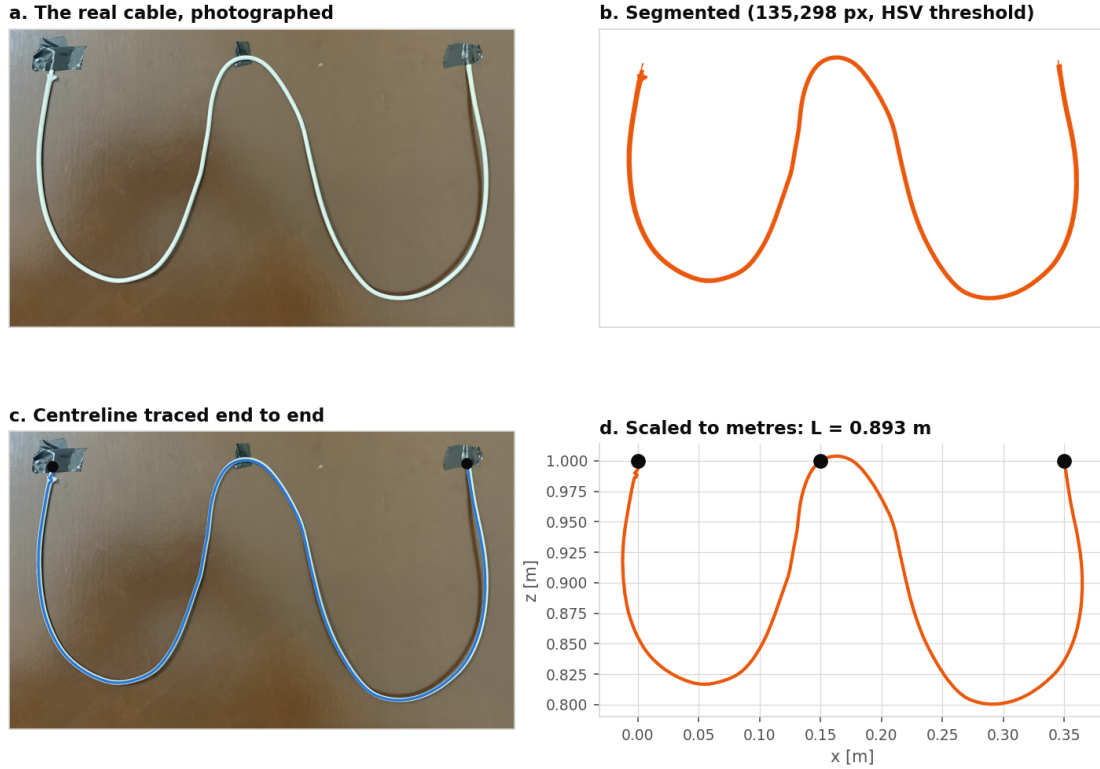


Figure 1: Where “real” comes from. The cable is photographed, segmented in HSV, traced along its own centreline, and scaled by the known clamp separation — giving a millimetre-accurate ground truth with no tape measure. Two independent photographs of this cable agree on its length to 0.32 mm.

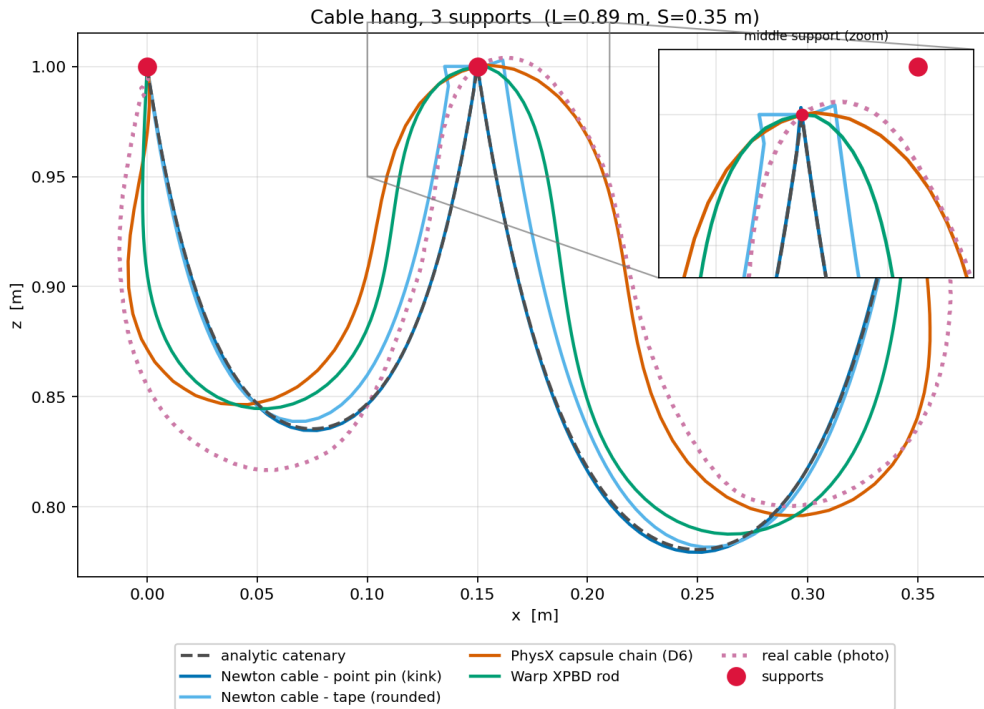


Figure 2: Every approach, the exact mathematics (dashed) and the real cable (dotted) on one axis. They bracket the real cable; the spread is boundary-condition modelling, not solver bugs.